

Employee Attrition Prediction using Machine Learning

Formative Assessment-02 · Data Analysis Case Study & Streamlit Deployment

Advanced Data Science [MCA33PE17] · MCA · SYMCA · Academic Year 2026-2027 · Semester I

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Live Streamlit app: [\[link to be added after deployment\]](#)

1. Problem and dataset

Goal: predict whether an employee will **leave the company** from HR data (role, pay, overtime, tenure, satisfaction, demographics) so HR can act *before* a resignation, and compare four classification algorithms.

Dataset: *IBM HR Analytics Employee Attrition & Performance* – a fictional sample dataset created by IBM data scientists (Kaggle / IBM GitHub): 1,470 employees, 30 usable features (23 numeric, 7 categorical), 237 leavers (16.1 %) and 1,233 stayers, no missing values.

Why this dataset: a real business problem where the two error types have different costs; mixed numeric and categorical data that needs genuine preprocessing (one-hot encoding, scaling, dropping constant / ID columns); strong class imbalance that makes accuracy misleading (always predicting “stays” gives 83.9 %); and a genuinely hard task where model choice, tuning and honest evaluation matter. Metrics are reported for the **leaving** class.

2. Exploratory data analysis - key insights

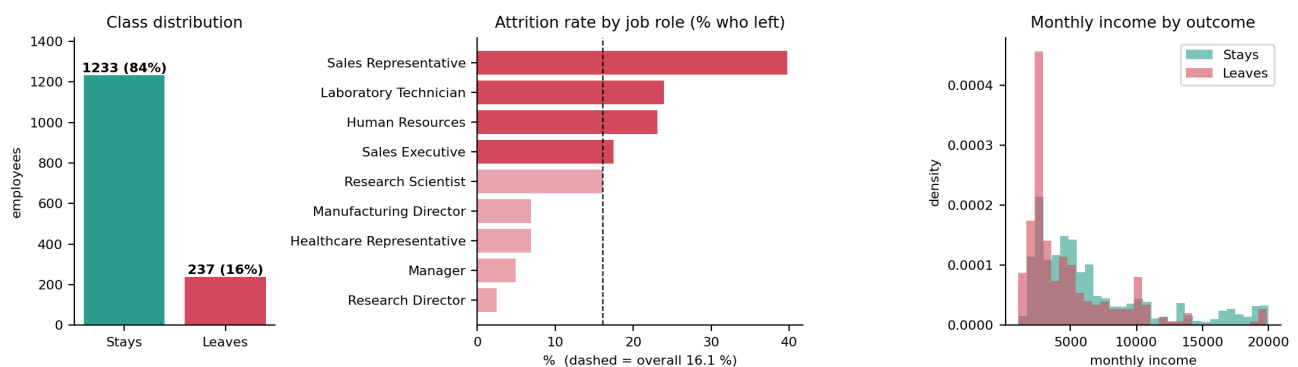


Figure 1 – class balance, attrition rate by job role, and income distribution of leavers vs. stayers.

- **Clean but imbalanced:** no missing values or duplicates; only 16.1 % leave. *EmployeeCount*, *StandardHours*, *Over18* are constant and *EmployeeNumber* is an ID – dropped.
- **Overtime is the strongest single signal:** 30.5 % of employees working overtime left vs. 10.4 % of the others; frequent travellers 24.9 % vs. 8.0 % (non-travel); single employees 25.5 % vs. 10-12.5 %.
- **Role and seniority:** Sales Representatives (39.8 %) and Laboratory Technicians (23.9 %) leave most; Managers (4.9 %) and Research Directors (2.5 %) least. No stock options: 24.4 %; Job Level 1: 26.3 %.
- **Leavers are younger, earlier-career and paid less:** median income 3,202 vs. 5,204, median age 32 vs. 36, median tenure 3 vs. 6 years.
- **No numeric feature is decisive:** all $|r|$ with attrition ≤ 0.17 (total working years -0.17 , job level -0.17 , monthly income -0.16) $\rightarrow$ a hard problem needing combinations of features. 7 of 253 numeric pairs have $|r| > 0.7$ (e.g. job level $\leftrightarrow$ income). Skewed features contain genuine (not erroneous) outliers, so they were kept.

3. Preprocessing

- **Columns:** 4 useless columns dropped; target encoded Yes $\rightarrow$ 1, No $\rightarrow$ 0. **Missing values:** none; median / most-frequent *SimpleImputer* kept as a safeguard for the deployed app.
- **Encoding:** 7 nominal columns **one-hot encoded** (51 model features in total) – label encoding would invent a false order between roles. **Scaling:** *StandardScaler* on numeric features for Logistic Regression and k-NN only (trees are scale-invariant).
- **No leakage:** everything sits in a *ColumnTransformer* inside a *Pipeline*, re-fitted in every CV fold. **Split:** stratified 80 % train (1,176) / 20 % test (294, 47 leavers), seed 42.

4. Models, cross-validation and tuning

Four algorithms were implemented with scikit-learn: **Logistic Regression, Decision Tree, Random Forest and k-NN**. Baselines were first checked with **stratified 5-fold cross-validation** on the training data (Table 1): the untuned Decision Tree and Random Forest reach 100 % training accuracy but only 79 % / 86 % in CV – clear over-fitting. **GridSearchCV** (same folds, score = F1 of the leaving class) was then applied to **all four** models; for LR and the trees the grid also tested `class_weight='balanced'`. The test set was never used to choose parameters.

Model	Train acc.	CV accuracy	CV F1 (leaves)	CV recall (leaves)
Logistic Regression	0.900	0.890 ± 0.019	0.561 ± 0.098	0.442 ± 0.096
Decision Tree	1.000	0.794 ± 0.017	0.356 ± 0.081	0.358 ± 0.102
Random Forest	1.000	0.863 ± 0.007	0.315 ± 0.078	0.200 ± 0.061
k-NN	0.876	0.852 ± 0.015	0.228 ± 0.126	0.142 ± 0.086

Table 1 – baseline 5-fold CV results (mean ± std).

Model	Best parameters (GridSearchCV)	Combos	Baseline CV F1	Tuned CV F1
Logistic Regression	{'C': 1, 'class_weight': None}	8	0.5610	0.5610
Decision Tree	{'class_weight': 'balanced', 'max_depth': 4, 'min_samples_leaf': 10}	48	0.3556	0.4288
Random Forest	{'class_weight': 'balanced', 'max_depth': 8, 'min_samples_leaf': 3, 'n_estimators': 100}	36	0.3153	0.4504
k-NN	{'n_neighbors': 3, 'p': 1, 'weights': 'uniform'}	24	0.2282	0.2798

Table 2 – GridSearchCV: tuning raised CV F1 for the three weaker models; Logistic Regression kept its defaults.

5. Results and model comparison (unseen test set, leaves = positive class)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Missed leavers	False alarms	Baseline F1
Logistic Regression	0.861	0.615	0.340	0.438	0.812	0.584	31	10	0.438
Decision Tree	0.728	0.301	0.532	0.385	0.704	0.332	22	58	0.343
k-NN	0.847	0.545	0.255	0.348	0.664	0.343	35	10	0.233
Random Forest	0.827	0.433	0.277	0.338	0.781	0.433	34	17	0.241

Table 3 – tuned models on the 294 test employees (47 leavers, default 0.5 threshold); best model highlighted.

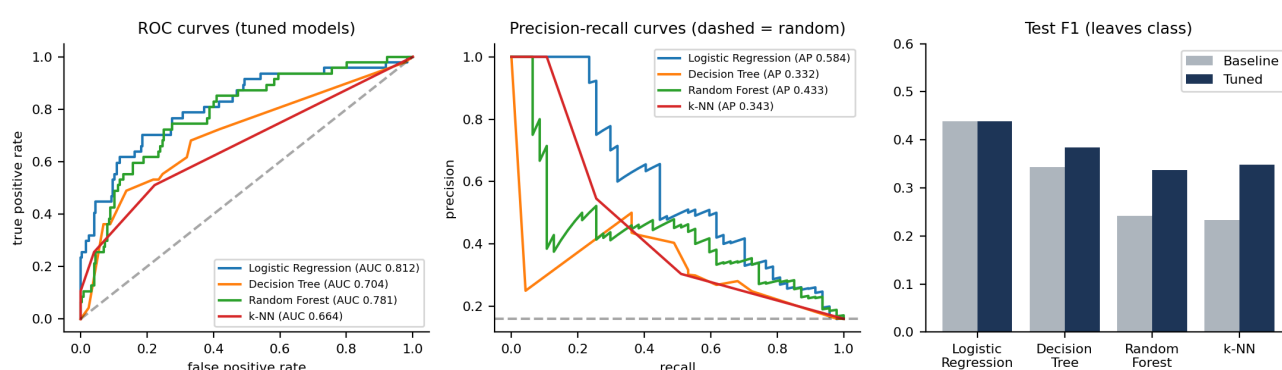


Figure 2 – ROC and precision-recall curves of the tuned models, and baseline-vs-tuned F1.

6. Insights and conclusions

- **Best model: Logistic Regression** – highest CV F1 (0.561), ROC-AUC 0.812 and PR-AUC 0.584 (random ≈ 0.16). The signal is mostly additive (overtime, travel, role, income, tenure), which a regularised linear model captures well from ~1,200 training rows.
- **Accuracy is misleading:** every model reaches 73–86 % accuracy, yet at the default 0.5 threshold the best one finds only 34 % of leavers.
- **Trees over-fit, tuning helps them most:** CV F1 rises 0.356 → 0.429 (Decision Tree), 0.315 → 0.450 (Random Forest), 0.228 → 0.280 (k-NN); winning grids use `class_weight='balanced'` and limited depth.

They still do not catch the linear model. k-NN is weakest: distances over 51 mostly one-hot features are dominated by noise.

- **Business view - the threshold matters:** lowering the alert threshold from 0.5 to 0.25 raises Logistic Regression's recall from 34 % to 62% (precision 50%, F1 0.55); HR would talk to 58 of 294 employees (20%) and catch 29 of 47 leavers.
- **Drivers:** overtime, frequent business travel, Laboratory Technician / Sales Representative roles and long time since promotion raise the risk; Research Director, no overtime, no travel and more total working years lower it – consistent with the EDA.
- **Limitations:** fictional single-snapshot data, only 47 test leavers (one employee $\approx$ 2 points of recall), associations rather than causes. **Not for real employment decisions.**

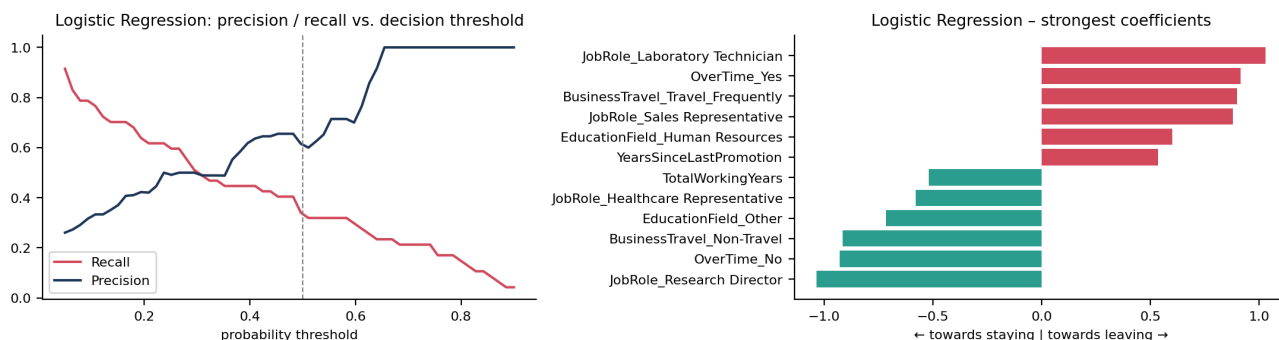


Figure 3 – precision / recall vs. decision threshold, and the strongest Logistic Regression coefficients.

7. Streamlit application (deployment)

`app.py` loads the four tuned pipelines saved with `joblib` and provides: (1) an input form with all 30 HR attributes grouped into Personal / Job / Pay & tenure / Satisfaction tabs (sliders, number boxes, drop-downs) with a **Predict** button and one-click examples (typical stayer / leaver, random real test records); (2) **every model's probability of leaving**, a consensus vote, an adjustable alert threshold and a “Why?” panel listing the factors that raise or lower the risk; (3) a **Model comparison** tab (metrics, confusion matrices, tuning and threshold tables computed live on the test split); (4) a **Dataset & insights** tab. Run locally with `streamlit run app.py`.

Live Streamlit app: [\[link to be added after deployment\]](#)

Educational demo on a fictional dataset – not for real employment decisions.